

# DAKSH SHARMA

## Software Engineer | Backend Architecture & AI Integrations

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### PROFILE SUMMARY

Innovative Full Stack Developer with advanced skills in the MERN stack and a targeted focus on building scalable backend infrastructures and mobile-first consumer experiences for the quick-service retail industry. Experienced in designing RESTful APIs, secure payment integrations, and deploying AI-driven models for customer retention and supply chain optimization. Currently expanding expertise in AWS and cloud-native deployments to ensure robust microservice scalability and zero downtime during high-traffic daily order surges.

### SKILLS:

- Frontend: React.js, Redux, Tailwind CSS, JavaScript, TypeScript
- Backend: Node.js, Express.js, FastAPI, Celery, Socket.io (WebSockets), REST, Server-Sent Events (SSE)
- Databases & Caching: PostgreSQL, TimescaleDB, MongoDB, Redis, Qdrant (Vector DB), SQL
- AI & Machine Learning: XGBoost, Scikit-learn, Ollama (Local LLMs), Groq, RAG Architecture, Python
- Cloud & DevOps: Docker, Kubernetes (K8s), AWS, Git/GitHub, Postman

### RELEVANT PROJECTS:

#### NexusRAG (AI ChatApp)

- Engineered a highly available RAG platform deployed on Kubernetes, featuring a hybrid search flow (Dense + BM25) and cross-encoder reranking for highly accurate retrieval.
- Optimized system performance by implementing SSE for real-time LLM token streaming and a background Redis pipeline to handle heavy document ingestion without blocking the API.
- **Tech:** FastAPI, React (Vite), MongoDB, Qdrant, Redis, Groq LLM, Docker, Kubernetes
- **GitHub:** <https://github.com/Daksh54/ChatBot>

#### TradeGenius (AI-Powered Algorithmic Trading Platform)

- Engineered an autonomous trading engine utilizing XGBoost and conformal prediction to generate risk-calibrated execution signals, successfully reducing maximum drawdown by over 50% (vs. benchmark) in historical backtesting.
- Architected a low-latency asynchronous pipeline with Celery and TimescaleDB, processing live market data with <300ms broker API execution latency via Fyers.
- Integrated a local LLM agent (Ollama) to automate strategy explainability and deployed a time-series database to optimize historical data querying and feature generation.
- **Tech:** Python, Node.js, React (TypeScript), XGBoost, TimescaleDB, Redis, Celery, Ollama, Docker, Upstox/Fyers API
- **GitHub:** [github.com/Daksh54/Trade\\_Genius.git](https://github.com/Daksh54/Trade_Genius.git)

#### RoastFlow (AI-Powered Retail Platform)

- Engineered a full-stack D2C platform featuring a Python/FastAPI microservice for ML-based product recommendations and a local LLM (Ollama) serving as an interactive "Roast Concierge."
- Optimized system performance by decoupling heavy tasks from the Node.js event loop and utilizing Redux Toolkit (RTK Query) state caching for a dynamic, low-latency "Coffee Exchange."
- **Tech:** MERN Stack, Python (FastAPI), Redux (RTK Query), Tailwind CSS, Ollama, Scikit-learn
- **GitHub:** [github.com/Daksh54/MERN-Stack-Ecommerce-Project](https://github.com/Daksh54/MERN-Stack-Ecommerce-Project)
- **Live:** [mern-stack-ecommerce-project-mocha.vercel.app](https://mern-stack-ecommerce-project-mocha.vercel.app)

#### ChatMERN (Real-Time Messaging Microservices)

- Engineered a distributed architecture integrating a Node.js/MongoDB core with a dedicated Python (FastAPI) NLP microservice to execute real-time sentiment analysis on live chat data.
- Architected a scalable, Kubernetes-deployed WebSocket backend, implementing an O(1) in-memory socket mapping system for instant, low-latency message broadcasting.
- **Tech:** React (Zustand), Node.js, MongoDB, Socket.io, Python (FastAPI), Docker, Kubernetes
- **GitHub:** [github.com/Daksh54/ChatMern.git](https://github.com/Daksh54/ChatMern.git)

### EDUCATION:

B.Tech – Computer Science & Information Technology Dronacharya College of Engineering, Gurgaon  
2023 – 2027

### CERTIFICATIONS

- Web Development Bootcamp [MERN Stack] – CodeHelp
- The Complete Full Stack Bootcamp (Udemy)
- Postman Student Expert